



CSE260: Digital Logic Design
BRAC University, Spring 2026
Submission of the Lab Report 02

Submitted by
Group 01
Section 03A

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Date of Submission – 27th April, 2026

Lab Report 2

Lab Experiment 4

Experiment name: Design and Implementation of 4-bit Parallel Binary Adder

Required components:

1. IC 7408
2. IC 7432
3. IC 7486
4. IC 7483

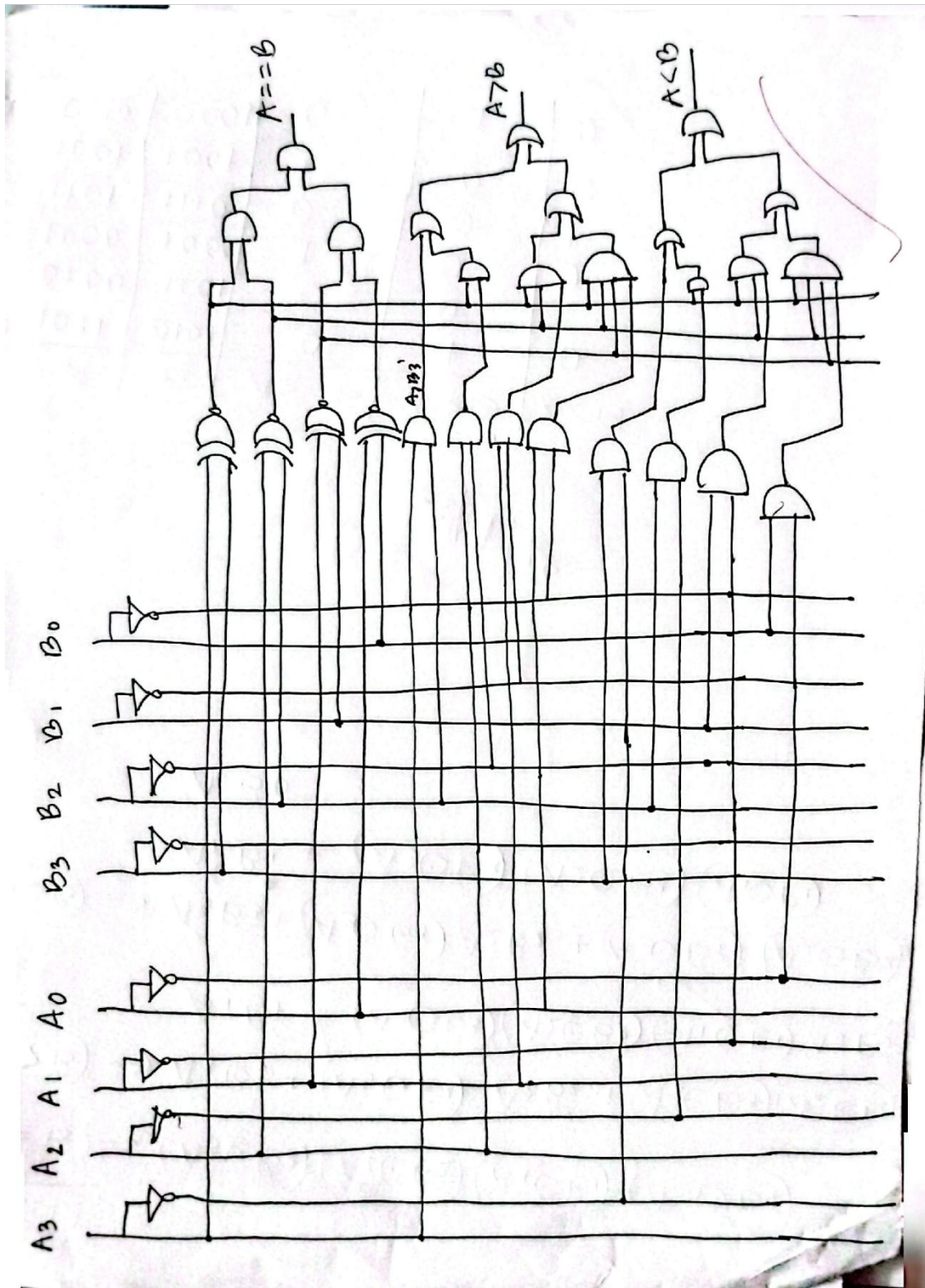
Lab Experiment 5

Experiment name: Implementation of 4-bit Magnitude Comparator

Required components:

1. IC 7408
2. IC 7432
3. IC 7404
4. IC 4077

Experimental Setup:



(4 bit)

$$(A=B) \rightarrow (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)(A_0 \odot B_0)$$

$$(A > B) \rightarrow A_3 B_3' + (A_3 \odot B_3) \cdot A_2 B_2' + (A_3 \odot B_3)(A_2 \odot B_2) A_1 B_1' + (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1) A_0 B_0'$$

$$(A < B) \rightarrow A_3' B_3 + (A_3 \odot B_3) A_2' B_2 + (A_3 \odot B_3)(A_2 \odot B_2) A_1' B_1 + (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1) A_0' B_0$$

Moham
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A	B	$A=B$	$A>B$	$A<B$
1011	0101	0	1	0
0100	1001	0	0	1
1000	1000	1	0	0
1101	1110	0	0	1
1001	1001	1	0	0
0010	0001	0	1	0

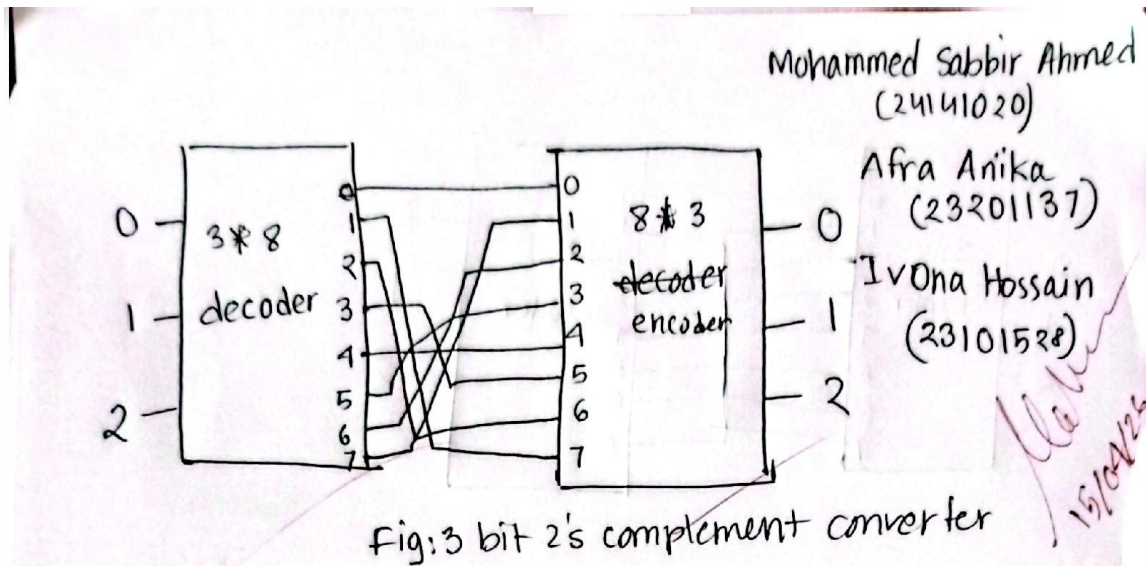
Lab Experiment 6

Experiment name: Design circuit using encoder & decoder.

Required components:

1. IC 74138
2. IC 74148

Experimental Setup:



Inputs				Expected outputs				Active low outputs		
minterm	C	B	A	minterm	D ₂	D ₁	D ₀	D ₂	D ₁	D ₀
0	0	0	0	0	0	0	0	1	1	1
1	0	0	1	7	1	1	1	0	0	0
2	0	1	0	6	1	1	0	0	0	1
3	0	1	1	5	1	0	1	0	1	0
4	1	0	0	4	1	0	0	0	1	1
5	1	0	1	3	0	1	1	1	0	0
6	1	1	0	2	0	1	0	1	0	1
7	1	1	1	1	0	0	1	1	1	0

Lab Experiment 7

Experiment name: Function Implementation Using a 4x1 MUX.

Required components:

1. IC 74153
2. IC 7408
3. IC 7432
4. IC 7404

Experimental Setup:

$F(A, B, C, D) = \sum(1, 2, 3, 4, 5, 6, 7, 9, 11, 12, 13)$

		I_0	I_1	I_2	I_3
00	$A'B'$	0	①	②	③
01	$A'B$	④	⑤	⑥	⑦
10	AB'	8	⑨	10	⑪
11	AB	⑫	⑬	14	15
		B	1	A'	$A'+B'$

$A'B + AB$
 $= B(A' + A) = B$

$A'B' + A'B$
 $= A'(B' + B)$
 $= A'$

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Discussion

[For discussion part, answer the following two questions only]

1. Construct an excess-9 to excess-5 converter using encoder and decoder.

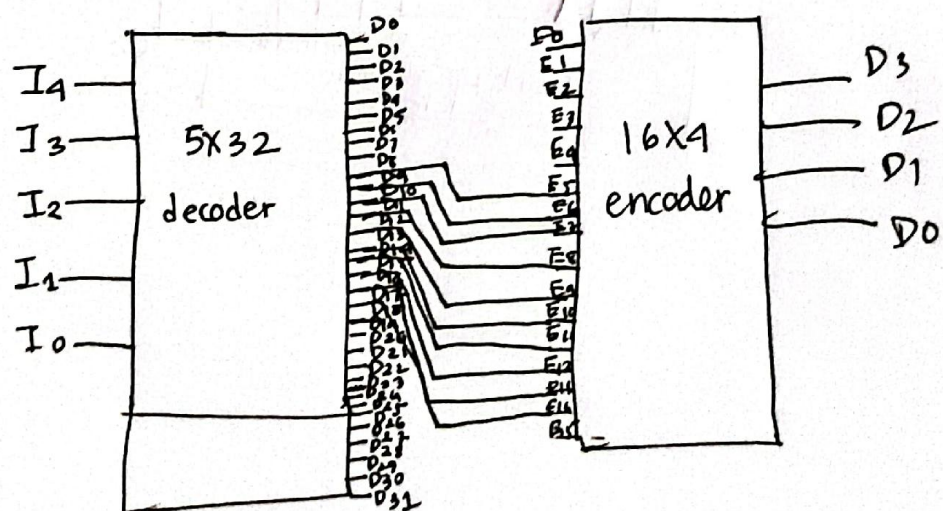
Discussion

1 e-9 to e-5 using encoder decoder

input range = 1001 - 10010 [used 5X32 decoder]

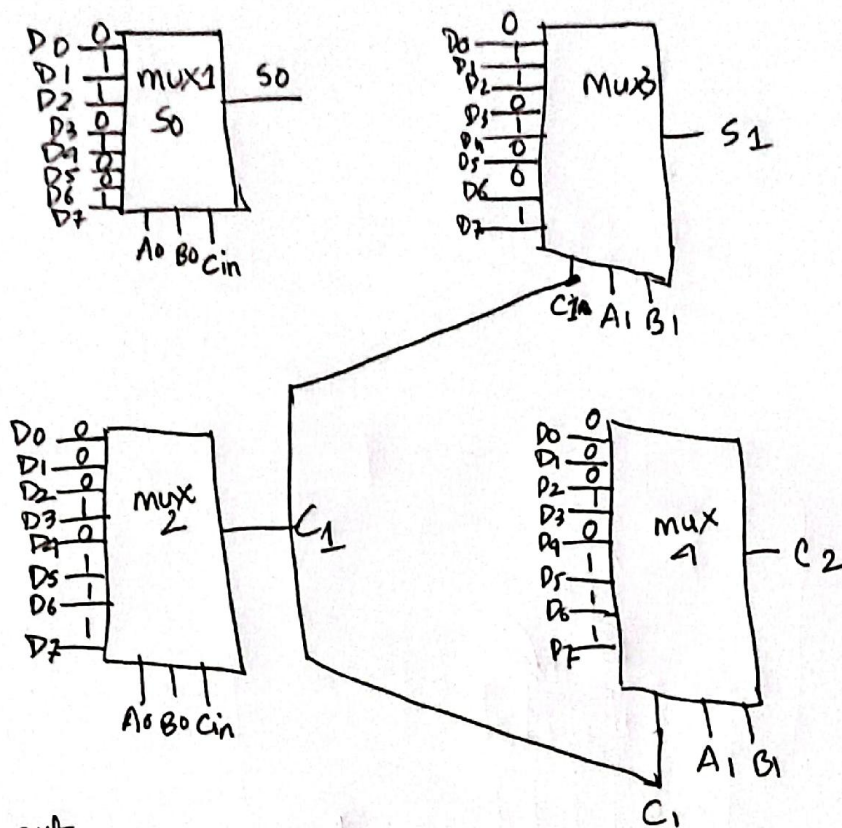
output n = 0101 - 1110 [used 16X4 encoder]

Decimal	Input (excess-9)	decoder output	connected encoder in	Output es
0	01001	D ₉	E ₅	0101
1	01010	D ₁₀	E ₆	0110
2	01011	D ₁₁	E ₇	0111
3	01100	D ₁₂	E ₈	1000
4	01101	D ₁₃	E ₉	1001
5	01110	D ₁₄	E ₁₀	1010
6	01111	D ₁₅	E ₁₁	1011
7	10000	D ₁₆	E ₁₂	1100
8	10001	D ₁₇	E ₁₃	1101
9	10010	D ₁₈	E ₁₄	1110



2. Design a 2-bit Parallel adder using exactly four 8:1 Mux(s)

2
2 bit parallel adder using 4 8X1 mux



Input

mux selector

A₀ B₀
A₁ B₁

Coded mux
datapin

D₀, D₁, D₂, D₄ → GND
D₃, D₅, D₇, D₆ → Power
+5V

Output

S₀ S₁,
C₁ C₀

